General Emergence Mechanics

Persistence, Coherence Shells, and the Three Axes of Stable Existence

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Abstract

This paper introduces *General Emergence Mechanics* (GEM), a framework in which emergence is understood as the observable outcome of persistence within a coherence shell. The observable universe is modelled as a dynamical coherence shell anchored at the 255° vacuum minimum. Stable structures and patterns arise and persist only when three coupled axes—shell depth, stiffness, and viscosity/relaxation time—jointly support low-gradient coherence. Within this framework, time is not fundamental but emerges as the directional relaxation of the shell toward its preferred minimum. GEM provides a mechanical language for persistence, stability, degradation, and failure across physical, biological, chemical, computational, and engineered systems.

This paper develops the Lagrangian foundation of GEM, derives the three axes and their coupling, and shows how time itself can be understood as emergent relaxation within the shell. It then examines the framework's consistency with known physical phenomena, including transient excitations, and its initial computational expression in engineered systems.

1 Introduction – The Problem of Emergence

The question of emergence—why stable structures, patterns, and "things" persist in the universe rather than dissolving into noise—remains one of the deepest open problems in physics and philosophy. Existing approaches commonly treat emergence either as a brute fact, an epiphenomenon of complexity, or a higher-level description that does not reduce cleanly to underlying dynamics. In each case, the mechanism by which a configuration becomes persistent rather than transient is left unexplained or deferred to statistical arguments.

General Emergence Mechanics (GEM) proposes a different approach. It treats the observable universe as a dynamical coherence shell anchored at the 255° vacuum minimum. Within this shell, stable existence is not assumed but arises as the consequence of three coupled dynamical axes: shell depth, stiffness, and viscosity/relaxation time. Together these determine whether a configuration can persist long enough to count as a stable entity.

The central claim of GEM is mechanical rather than philosophical: *emergence is governed by persistence, and persistence is governed by the shell's dynamical properties.*

2 The Lagrangian Foundation of GEM

The foundation of GEM is a modified Lagrangian density that extends the existing MT/GCF action with terms that define the coherence shell and its dynamical axes. The complete action is written schematically as

$$\mathcal{L}_{\mathrm{GEM}} = \mathcal{L}_{\mathrm{MT}} + \mathcal{L}_{\mathrm{shell}} + \mathcal{L}_{\mathrm{persistence}}. \tag{1}$$

Here $\mathcal{L}_{\mathrm{MT}}$ denotes the underlying Modal Theory / Global Coherence Framework action containing the 255° vacuum minimum, orbifold structure, bias term, and decoherence dynamics.

2.1 Coherence Shell Potential

We introduce an effective basin-shaped potential centered on the 255° vacuum minimum $\phi_* = 255°$:

$$V(\phi) = \frac{\kappa}{2}(\phi - \phi_*)^2 + \frac{\alpha}{4}(\phi - \phi_*)^4. \tag{2}$$

This potential is to be understood as the *effective expansion of the MT vacuum structure around the* 255° *minimum*, not as an independent second minimum. It creates the local energy basin that defines the coherence shell. The parameter κ sets the curvature near equilibrium, while α controls the higher-order shape of the basin.

2.2 Three-Axis Shell Dynamics

The shell's dynamical response is governed by three explicit contributions:

$$\mathcal{L}_{\text{shell}} = \frac{\beta}{2}(\phi - \phi_*)^2 + \frac{\lambda}{2}(\nabla\phi)^2 + \frac{\gamma}{2}(\partial_t\phi)^2 + \delta(\phi - \phi_*)^2(\nabla\phi)^2. \tag{3}$$

These terms encode the three axes:

- **Depth** (β): the effective energetic cost of finite local displacement within the occupied shell regime.
- **Stiffness** (λ): the penalty on spatial phase gradients.
- **Viscosity / Relaxation** (γ): the coarse-grained temporal response that sets the characteristic recovery timescale.

A useful effective timescale is

$$\tau \sim \frac{\gamma}{\lambda}, \tag{4}$$

which expresses the relative balance between temporal drag and spatial restoring force.

The term proportional to δ is a simple coupling term. It ensures that depth and stiffness do not act independently, but interact through the local phase configuration.

2.3 Persistence Engine Term

To bias the dynamics toward configurations capable of persistence, we add a phase-gradient coherence term:

$$\mathcal{L}_{\text{persistence}} = -\eta\left[1 - \exp\left(-\frac{|\nabla\phi|^2}{\epsilon^2}\right)\right], \qquad \eta > 0. \tag{5}$$

Here η sets the strength of the persistence bias and ϵ defines the gradient scale below which configurations are regarded as coherent. This term acts as a soft filter: smooth, low-gradient configurations are energetically favoured, while high-gradient regions are penalised but not forbidden.

2.4 Complete GEM Lagrangian

The full effective Lagrangian density is therefore

$$\mathcal{L}_{\text{GEM}} = \mathcal{L}_{\text{MT}}(\phi) + \frac{\beta}{2}(\phi - \phi_*)^2 + \frac{\lambda}{2}(\nabla\phi)^2 + \frac{\gamma}{2}(\partial_t\phi)^2 + \delta(\phi - \phi_*)^2(\nabla\phi)^2 - \eta\left[1 - \exp\left(-\frac{|\nabla\phi|^2}{\epsilon^2}\right)\right]. \tag{6}$$

2.5 Equations of Motion

Varying the action yields, schematically,

$$\gamma\,\partial_t^2\phi - \lambda\,\nabla^2\phi + \frac{\partial V_{\mathrm{MT}}}{\partial\phi} + \beta(\phi-\phi_*) + \text{coupling terms} + \text{persistence terms} = 0. \tag{7}$$

This field equation encodes restoring force toward the 255° minimum, spatial smoothing through stiffness, coarse-grained temporal response, and nonlinear bias toward low-gradient coherence.

At this stage, the viscosity contribution should be interpreted as an *effective coarse-grained temporal term*. A fully dissipative treatment would require a more complete non-conservative extension, for example through a Rayleigh-type dissipation functional.

3 Stability and Persistence in the Coherence Shell

Once the GEM Lagrangian is defined, the stability of the coherence shell and the conditions for persistence can be analysed by studying small perturbations about the vacuum minimum. Let

$$\phi(x,t) = \phi_* + \delta\phi(x,t), \qquad \phi_* = 255^\circ,$$

with

$$|\delta\phi| \ll 1$$

in the linear regime.

3.1 Linearised Shell Dynamics

To leading order, the effective shell dynamics are governed by the quadratic terms

$$\mathcal{L}_{\mathrm{lin}} = \frac{\gamma}{2}(\partial_t\delta\phi)^2 - \frac{\lambda}{2}(\nabla\delta\phi)^2 - \frac{\kappa}{2}(\delta\phi)^2.$$

The corresponding linearised equation of motion is

$$\gamma\,\partial_t^2\delta\phi - \lambda\,\nabla^2\delta\phi + \kappa\,\delta\phi = 0.$$

For Fourier modes

$$\delta\phi(x,t) \propto e^{i(k\cdot x-\omega t)},$$

this yields the dispersion relation

$$\gamma\omega^2 = \kappa + \lambda k^2.$$

The shell is therefore linearly stable provided

$$\kappa > 0, \qquad \lambda > 0, \qquad \gamma > 0,$$

since these conditions ensure

$$\omega^2 > 0$$

for all wavevectors k.

3.2 Correlation Length and Spatial Persistence

The competition between the local restoring term and the gradient penalty defines a characteristic correlation length

$$\xi = \sqrt{\frac{\lambda}{\kappa}}.$$

This is the natural spatial persistence scale of the shell. Perturbations on scales much larger than ξ remain coherent more easily, while perturbations on scales much smaller than ξ are strongly suppressed. The corresponding critical wavelength is

$$L_c \sim 2\pi\xi = 2\pi\sqrt{\frac{\lambda}{\kappa}}.$$

3.3 Large-Perturbation Confinement

Beyond the linear regime, the quartic correction in the shell potential becomes important:

$$V(\phi) = \frac{\kappa}{2}(\phi - \phi_*)^2 + \frac{\alpha}{4}(\phi - \phi_*)^4.$$

The restoring force is

$$F_{\text{restore}} = -\frac{\partial V}{\partial \phi} = -\kappa\Delta\phi - \alpha(\Delta\phi)^3, \qquad \Delta\phi = \phi - \phi_*.$$

The quartic term preserves confinement of the shell basin so long as the perturbation remains within the effective bound

$$|\Delta\phi| \lesssim \sqrt{\frac{\kappa}{\alpha}}.$$

This defines the amplitude scale beyond which the perturbation is no longer well described as confined within the local shell basin.

3.4 Effective Relaxation and Persistence Threshold

In the full GEM picture, persistence requires more than linear stability: a configuration must remain coherent for longer than the shell's characteristic recovery time. At the effective level, this may be expressed as

$$T_{\text{persist}} > \tau_{\text{shell}}.$$

A fully dissipative treatment would require a non-conservative extension, for example through a Rayleigh-type dissipation term. In such a treatment, the persistence threshold separates transient excitations from stable entities by comparing the excitation lifetime to the shell relaxation timescale.

3.5 Coherence Shell Stability Theorem (Effective Form)

Given positive parameters

$$\kappa, \lambda, \gamma, \alpha > 0,$$

the coherence shell centred at

$$\phi_* = 255^\circ$$

is linearly stable when:

- $\kappa > 0$ ensures a local restoring minimum,
- $\lambda > 0$ suppresses uncontrolled spatial distortion,

- $\gamma > 0$ gives well-defined temporal response,
- $|\Delta\phi| \lesssim \sqrt{\kappa/\alpha}$ keeps perturbations within the effective shell basin.

Under these conditions, perturbations remain bounded and the shell supports coherent structures on spatial scales larger than the characteristic correlation length

$$\xi = \sqrt{\frac{\lambda}{\kappa}}.$$

3.6 Implications

The shell therefore exhibits scale-dependent persistence. Long-wavelength structures (low k) are more readily supported, while short-wavelength fluctuations (high k) are suppressed. Depth sets the energetic barrier to escape, stiffness controls spatial smoothing, and viscosity/relaxation time governs the temporal character of recovery. Together they define the persistence conditions under which stable entities can emerge.

4 Time as Emergent Relaxation

Within GEM, time is not treated as a fundamental backdrop in which the coherence shell exists. Rather, it emerges from the shell's own dynamics, with its temporal character set by the viscosity/relaxation axis.

A local perturbation displaces the vacuum phase from the 255° minimum,

$$\phi(x,t) = \phi_* + \delta\phi(x,t), \qquad \phi_* = 255^\circ.$$

Because the shell possesses finite viscosity, the perturbed phase does not return instantaneously to equilibrium. Instead, it relaxes toward the minimum over a finite timescale. In the effective shell description, this characteristic recovery time is

$$\tau_{\text{shell}} \sim \frac{\gamma}{\lambda},$$

where γ sets the temporal response and λ sets the spatial restoring scale.

The finite interval required for this return to equilibrium generates the distinction between "before" and "after." Temporal direction is therefore not imposed externally, but emerges from the shell's irreversible tendency to relax toward its preferred minimum. In this sense, the arrow of time is the lived signature of finite-viscosity relaxation inside the coherence shell.

The **past** consists of those distortions that have already relaxed enough to leave a lasting record in the shell's configuration. The **present** is the instantaneous local posture of the shell—the current phase state. The **future** is the near-field of dynamically favoured but not yet completed relaxations: the paths the shell is gently pulling the system toward.

The limiting cases clarify the role of viscosity. In the limit

$$\gamma \to 0,$$

the shell would snap back instantaneously; there would be no temporal thickness, only reversible fluctuation. In the opposite limit

$$\gamma \to \infty,$$

the shell would remain permanently distorted and relaxation would cease; temporal unfolding would effectively freeze. Finite positive viscosity therefore produces the familiar unidirectional flow in which structures persist long enough to be experienced as stable.

More generally, if the phase displacement decays schematically as

$$\delta\phi(t) \sim \delta\phi_0\, e^{-t/\tau_{\text{shell}}},$$

then the shell possesses a preferred relaxation direction: perturbation amplitudes decrease monotonically toward the equilibrium state ϕ_*. Temporal asymmetry is thus the natural consequence of living inside a dynamical coherence shell with a preferred resting posture and a finite recovery time.

5 Virtual Particles and the Limits of Persistence

GEM provides a mechanical account of the distinction between transient excitations and stable entities. Within the coherence shell, the vacuum phase remains anchored near the 255° minimum. Local perturbations can briefly displace the phase from this preferred posture, giving rise to short-lived excitations. However, not every excitation is capable of persistence. Only configurations that remain sufficiently phase-coherent with the shell's dynamical conditions can survive long enough to stabilise.

In this framework, virtual particles are interpreted as transient distortions of the vacuum phase. They arise when a local perturbation momentarily displaces the phase, but fail to satisfy the full persistence conditions imposed by the shell. Because the shell has finite depth, stiffness, and relaxation time, such excitations are driven back toward the vacuum before they can acquire stable identity.

The coherence shell therefore imposes a persistence threshold. Configurations that remain within the shell's basin and maintain sufficient coherence for longer than the characteristic relaxation timescale can stabilise as durable entities, including particles, atoms, and higher-order structures. Configurations that fall short of this threshold remain transient. The distinction between a virtual excitation and a stable particle is thus not one of kind, but of persistence.

At the effective level, this criterion may be written as

$$T_{\text{persist}} > \tau_{\text{shell}},$$

where T_{persist} is the coherence lifetime of the excitation and τ_{shell} is the characteristic shell relaxation time. Excitations with

$$T_{\text{persist}} < \tau_{\text{shell}}$$

remain transient and are experienced as virtual, while excitations satisfying the persistence condition can acquire stable identity.

This provides a clean mechanical criterion for existence. The shell acts as a natural filter: only sufficiently coherent, long-lived configurations survive to become observable "things." Short-lived distortions remain real at the level of local phase displacement, but they do not persist long enough to become stable components of the observable world.

6 Applications and Testable Predictions

GEM is intended not only as a theoretical framework but as a source of concrete, testable predictions across multiple domains. Its three coupled axes—depth, stiffness, and viscosity/relaxation time—together with the persistence threshold provide a common mechanical language for stability, resilience, degradation, and failure.

6.1 Materials and Viscoelastic Systems

In materials science, GEM maps naturally onto viscoelastic behaviour. Shell depth corresponds to the energy barrier that must be overcome for permanent deformation or fracture. Stiffness

corresponds to the immediate elastic resistance of the material under load. Viscosity, or relaxation time, governs the rate at which stress relaxes, creep develops, or residual strain is retained.

This mapping suggests a clear set of predictions. Materials with high depth, high stiffness, and low viscosity should exhibit rapid recovery, low permanent deformation, and strong resistance under cyclic loading. Materials with high viscosity should instead show delayed recovery, long-term creep, hysteresis, or memory effects under repeated stress. Materials with low depth may appear locally stable under small loads but should fail abruptly once a critical threshold is crossed.

At the level of measurable quantities, one may write schematically

$$E_{\text{eff}} \sim \text{stiffness}, \qquad \tau_{\text{relax}} \sim \text{viscosity}, \qquad \Delta E_{\text{yield}} \sim \text{depth},$$

so that persistence under cyclic stress is expected to depend on the joint condition

$$\Delta E_{\text{yield}} \gg k_B T_{\text{eff}}, \qquad \tau_{\text{relax}} \ll T_{\text{load}},$$

where T_{load} is the characteristic loading timescale. These predictions can be tested directly using standard rheological, creep, and fatigue measurements.

6.2 Biological and Chemical Homeostasis

In biological and chemical systems, GEM provides a mechanical language for resilience, regulation, and recovery. Cellular stress responses, immune regulation, metabolic control, and chemical buffering can all be interpreted as processes that maintain the local system near a stable persistence basin. Depth determines how far the system can be driven before irreversible breakdown or transition occurs. Stiffness governs the immediate buffering capacity against disturbance. Viscosity or relaxation time governs how quickly recovery occurs, and whether perturbations resolve cleanly or linger as chronic distortions.

GEM therefore predicts that biologically robust systems should display a favourable combination of sufficient depth, adequate stiffness, and moderate-to-low effective viscosity. Systems with high viscosity and reduced stiffness should be more prone to prolonged dysregulation, delayed recovery, or chronic pathological states. In chemical systems, similar distinctions should appear in buffering, equilibration, and recovery from perturbation.

At the schematic level, one may express resilience by a persistence inequality of the form

$$T_{\text{recovery}} < T_{\text{damage}},$$

where T_{recovery} is the characteristic relaxation time of the system and T_{damage} is the timescale on which irreversible injury or transition occurs. These predictions can, in principle, be explored in controlled models of inflammation, stress adaptation, metabolic recovery, and chemically buffered reaction systems.

6.3 Computational Systems: Frankie as a Case Study

The Frankie v5–v7 architecture provides an initial computational case study for GEM-like dynamics. High-level external routing acts as a coarse coherence controller, while internal mechanisms such as phase-weighted attention, cache persistence shaping, and hidden-state evolution act as low-level persistence biases. Together they implement a selective persistence regime: costly operations are preferentially invoked when conversational state becomes unstable, high-tension, or analytically demanding, while stable low-gradient cases are handled with reduced expenditure.

Across benchmark runs, these mechanisms produced measurable efficiency improvements relative to a monolithic baseline, including substantial reductions in token usage ($\approx$ 30–37%),

latency ($\approx$ 40–47%), and overall work ($\approx$ 22–27%). The precise gains varied across versions and test sets, but the overall pattern remained stable: selective persistence and coherence-aware control improved computational efficiency without collapsing output quality.

A schematic computational analogue of the persistence condition may be written as

$$C_{\text{state}} > C_{\min} \quad \Longrightarrow \quad \text{cheap stable processing,}$$

$$C_{\text{state}} < C_{\min} \quad \Longrightarrow \quad \text{costly corrective processing,}$$

where C_{state} denotes an effective coherence measure of the active conversational or representational state. Frankie therefore serves as an engineered proof-of-concept that GEM-style dynamics can be implemented, measured, and iteratively refined in practice.

In this way, GEM does not remain confined to speculative physical interpretation. It also offers a transferable mechanical framework for analysing and designing systems whose central problem is not mere computation, but persistent organised behaviour under stress.

7 Relation to MT/GCF and Broader Implications

General Emergence Mechanics (GEM) is not a separate theory from Modal Theory / the Global Coherence Framework (MT/GCF), but a direct development of it. MT/GCF provides the underlying vacuum structure: the 255° minimum, the orbifold geometry, the bias term, and the decoherence dynamics. GEM extracts the mechanical consequences of that structure for persistence, stability, and emergence.

Within this picture, the coherence shell is the natural basin formed around the 255° vacuum lock. The three axes—depth, stiffness, and viscosity/relaxation time—together with the persistence threshold, are not introduced as new fundamental postulates. Rather, they are the dynamical properties that arise once the vacuum is treated as a phase-stable basin with finite resistance to distortion and finite recovery time. In this sense, GEM is the operational mechanics of MT/GCF as applied to the question of why stable structures exist at all.

The relation may be expressed schematically as

$$\text{MT/GCF} \Longrightarrow \text{vacuum structure,} \qquad \text{vacuum structure} \Longrightarrow \text{coherence shell,}$$

$$\text{coherence shell} \Longrightarrow \text{persistence conditions,}$$

so that GEM functions as the effective mechanical layer through which MT/GCF is connected to the observable world of stable entities.

GEM therefore bridges the gap between MT/GCF's geometric and field-theoretic description and the observable world of persistent entities. It proposes that the same vacuum principles that govern the structure of the underlying field also govern the conditions under which any configuration can persist long enough to acquire stable identity. GEM is thus both a refinement of MT/GCF and an extension of it into a broader mechanics of emergence.

The implications of this extension are not confined to physics. In information theory, GEM suggests a mechanical basis for memory, stability, and information preservation. In biology and chemistry, it reframes homeostasis, resilience, and recovery in terms of shell dynamics. In computation, it offers an explanation for why selective persistence, as demonstrated in the Frankie architecture, can reduce cost without sacrificing capability. In engineered systems more generally, it suggests design principles based on coherence, recovery, and persistence rather than on brute-force optimisation alone.

At the level of shared form rather than shared substrate, GEM proposes a transferable persistence structure of the schematic kind

$$\text{stability} \sim f(\text{depth}, \text{stiffness}, \text{viscosity}, \text{persistence threshold}),$$

with different domains realising the same dynamical pattern in different physical, biological, chemical, or computational media. In this sense, GEM does not require that all systems share the same microscopic ontology; it requires only that they exhibit analogous persistence mechanics.

By providing a common language of persistence across domains, GEM opens the possibility of genuine cross-disciplinary transfer. Concepts, models, and experimental techniques developed in one field may illuminate others when translated into the language of shell depth, stiffness, viscosity, and persistence thresholds. In this framework, the coherence shell is not merely metaphorical. It is the central dynamical object through which emergence, stability, and failure can be described in unified mechanical terms.

8 Relation to MT/GCF and Broader Implications

8.1 Effective Reduction from MT/GCF to GEM

GEM is not an independent theory but an effective reduction of the full MT/GCF vacuum dynamics to the mechanics of persistence:

$$\mathcal{L}_{\text{GEM}} = \mathcal{R}[\mathcal{L}_{\text{MT/GCF}}]$$

where $\mathcal{R}$ denotes the reduction map that expands the MT vacuum potential around the 255° minimum, coarse-grains temporal response into an effective viscosity γ, and projects the full field dynamics onto the three coupled axes (depth, stiffness, relaxation). This reduction preserves the essential constraint structure of MT while focusing on the observable signature of persistence.

8.2 Cross-Domain Persistence Functional

Across physical, biological, chemical, computational, and engineered systems, persistence can be expressed through a common functional form:

$$P(D, S, \tau) = \frac{D \cdot S^{-1} \cdot \tau}{\Theta}$$

where D is effective depth (energy barrier), S is stiffness (resistance to deformation), τ is relaxation time (recovery rate), and Θ is a domain-specific normalization constant. Configurations persist when

$$P(D, S, \tau) > P_{\min}.$$

8.3 Universality of the Three-Axis Coupling

The coupling relation

$$\frac{\kappa}{\lambda} \cdot \tau = \text{constant}$$

is universal across domains, although the physical interpretation of κ, λ, and τ varies. The following table illustrates the correspondence:

Domain	Depth (κ)	Stiffness (λ)	Relaxation time (τ)
Materials	Deformation barrier	Elastic modulus	Stress-relaxation time
Biology	Homeostatic range	Regulatory stiffness	Recovery timescale
Computation	State-space basin	Attention resistance	Convergence time
Physics	Vacuum curvature	Gradient penalty	Decoherence time

This structural isomorphism allows GEM to serve as a transferable language for persistence, stability, and failure across disciplines.

8.4 Bridge to Standard Physics

GEM's persistence functional provides a quantitative bridge to established physics:

$$P_{\text{physical}}(D, S, \tau) = \frac{\xi^2}{\tau_{\text{relax}}}$$

where $\xi = \sqrt{\lambda/\kappa}$ is the correlation length and $\tau_{\text{relax}} = \gamma/\lambda$ is the relaxation time. Known coherence lengths (superconductivity, superfluidity, liquid crystals) correspond to GEM's persistence scale ξ; measured relaxation times map directly to τ.

8.5 Implications for a Theory of Everything

If GEM's persistence functional $P(D, S, \tau)$ is universal, then:

- Unification: Different domains share the same dynamical form for stability.
- Reduction path: Full MT/GCF $\rightarrow$ effective GEM $\rightarrow$ domain-specific laws.
- Testability: The same inequality $P > P_{\min}$ predicts persistence in materials, biology, computation, and cosmology.

This suggests that persistence mechanics may be a fundamental layer beneath domain-specific laws — not replacing them, but explaining why they support stable entities.

9 Conclusion

GEM shifts the explanatory focus from geometry alone to persistence mechanics:

- Emergence becomes persistence under constraints.
- Time emerges as directional relaxation in a coherence shell.
- Reality consists of persistent configurations satisfying $P(D, S, \tau) > P_{\min}$.

The framework is mathematically explicit ($\mathcal{L}_{\text{GEM}} = \mathcal{R}[\mathcal{L}_{\text{MT/GCF}}]$), testable ($P > P_{\min}$), and cross-domain ($P(D, S, \tau)$ universal). Next steps include deriving domain-specific forms of P, measuring ξ and τ experimentally in materials and biological systems, and extending the framework to computational architectures such as Frankie.

GEM offers a transferable mechanical language for emergence, stability, and failure — turning what was previously descriptive geometry into a predictive theory of persistence.